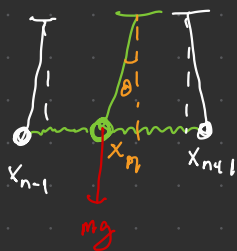


- Calculate the **equation of motion of the n th** pendulum describing small horizontal oscillations.
- Calculate and plot the **dispersion relation** for this coupled system.
- Calculate the **maximum phase velocity** and **minimum group velocity** on this dispersion curve
- For the **lowest and highest available wave numbers** sketch the pendulum configurations.

Q. (I don't know the part c)

a-



$$T_{n-1} = k(x_{n-1} - x_n) \quad T_{n+1} = k(x_{n+1} - x_n)$$

$$T_{\text{spring}} = k(x_{n-1} + x_{n+1} - 2x_n)$$

$$T_{\text{pendulum (horizontal)}} = -mg \cdot \frac{x_n}{L}$$

$$\frac{T}{m} = a = \ddot{x}_n = \frac{k}{m}(x_{n-1} + x_{n+1} - 2x_n) - \frac{g}{L}x_n$$

$$x_0 = \psi(0, t) = 0 = \cos(\omega t + \phi) [0 + B] \rightarrow B = 0$$

$$x_{10} = \psi(1, t) = 0 = \cos(\omega t + \phi) [A \sin(\frac{2\pi}{\lambda}) + 0] \rightarrow \frac{2\pi}{\lambda} = \pi, 2\pi, 3\pi, \dots$$

$$x_n = \psi(\frac{n}{10}, t) = A \cos(\omega t + \phi) \cdot \sin(k \frac{n}{10})$$

$$x_{n-1} = \psi(\frac{n-1}{10}, t) = \cos(\omega t + \phi) \cdot A \cdot \sin(\frac{k(n-1)}{10})$$

$$x_{n+1} = \psi(\frac{n+1}{10}, t) = \cos(\omega t + \phi) A \sin(\frac{k(n+1)}{10})$$

$$x_{n-1} + x_{n+1} = A \cos(\omega t + \phi) \left[\sin(\frac{k(n-1)}{10}) + \sin(\frac{k(n+1)}{10}) \right]$$

$$= \sin(\frac{kn}{10}) \cdot \cos(\frac{k}{10}) \cdot 2$$

$$x_{n-1} + x_{n+1} = 2x_n \cdot \cos(\frac{k}{10})$$

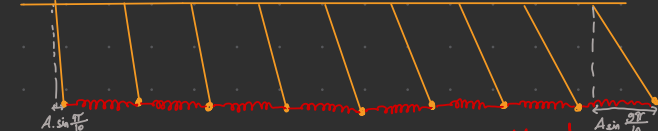
$$\ddot{x}_n = \frac{k}{m} \left(2x_n \cos \frac{k}{10} - 1 \right) - \frac{g}{L} x_n = x_n \left(2 \frac{k}{m} \left(\cos \frac{k}{10} - 1 \right) - \frac{g}{L} \right) = \ddot{x}_n$$

$$\omega = \sqrt{\frac{g}{L} + 2 \frac{k}{m} (1 - \cos \frac{k}{10})} = \sqrt{10 + 2 - 2 \cos \frac{k}{10}} = \sqrt{102 - 2 \cos \frac{k}{10}} = \omega$$

$k = \pi$ (d) lowest wave number

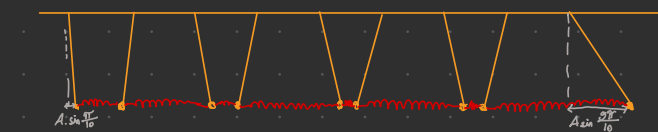
$$x_n = A \cos(\omega t + \phi) \cdot \sin(\frac{n\pi}{10})$$

$$t = 2\pi - \frac{\phi}{\omega}$$



$k = 9\pi$ (d) Highest wave number

$$x_n = A \cos(\omega t + \phi) \cdot \sin(\frac{9n\pi}{10})$$



$$\frac{\partial^2 \psi(z, t)}{\partial z^2} = \frac{T_0}{\rho} \frac{\partial^2 \psi(z, t)}{\partial t^2}$$

$$\psi(z, t) = A_{12} \cos(\omega t + \phi)$$

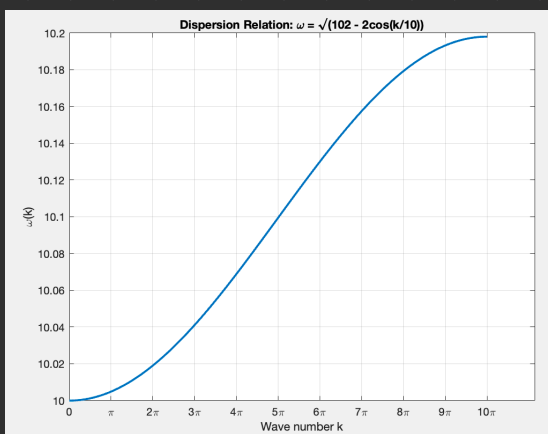
$$A(z) = A \sin(2\pi \frac{z}{\lambda}) + B \cos(2\pi \frac{z}{\lambda})$$

$$\psi(z, t) = \cos(\omega t + \phi) \left[A \sin(2\pi \frac{z}{\lambda}) + B \cos(2\pi \frac{z}{\lambda}) \right]$$

$$\frac{k}{m} = 1, \quad g = 10, \quad L = 0.1$$

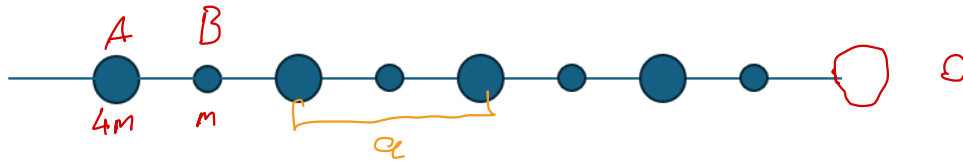
$$\ddot{x}_n = \left(2 \cos \frac{k}{10} - 102 \right) x_n$$

$$k \in \{ \pi, 2\pi, 3\pi, 4\pi, 5\pi, 6\pi, 7\pi, 8\pi, 9\pi \}$$



Dispersion Plot (b)

Q2. Consider a one-dimensional diatomic chain with lattice constant a , as shown in the figure. Each unit cell contains two atoms, A and B, with masses $M_A=4M_B$. Assume that there are five pairs of unit cells attached to a string under tension T_0 . $4m$ m



- Derive the **equations of motion** for the n th pair of atoms, describing **small verticle oscillations**.
- Determine and **plot the dispersion relation** for this coupled system (5 pairs of unit cells).
- Write a **MATLAB code** to visualize the **eigenmodes of the vertical oscillations**. Plot the **mode shapes** and calculate the **eigenfrequencies** corresponding to each mode.

You can use the attached code written for a chain of identical beads:

```
clear;

L=1;
numberOfBeads=5;
modeToDisplay=1;

a=L/(numberOfBeads+1);
for n=1:numberOfBeads
    for mode=1:numberOfBeads
        psi(mode,n)=1*sin((mode*pi/L)*n*a);
        w(mode)=(2/a)*1*sin(mode*pi*a/(2*L));
    end
end

for t=0:0.01:10

    scatter(L*(1:numberOfBeads)/(numberOfBeads+1),psi(modeToDisplay,:)*cos(w(modeToDisplay)*t),20,'filled');
    hold on;
    plot(L*(0:(numberOfBeads+1))/(numberOfBeads+1),[0
psi(modeToDisplay,:)*cos(w(modeToDisplay)*t) 0]);
    hold off;
    axis([0 L -1 1]);
    pause(0.001);
end
```

Q2

$$M_A \ddot{\psi}_n^A = T_0 \frac{\overbrace{(\psi_{n-1}^B - \psi_n^A)}^{\sin}}{a} + T_0 \frac{(\psi_n^B - \psi_n^A)}{a}$$

$$\psi_n^A = A \cos(\omega t + \varphi) \cdot \sin(kna)$$

$$M_B \ddot{\psi}_n^B = T_0 \frac{(\psi_n^A - \psi_n^B)}{a} + T_0 \frac{(\psi_{n+1}^A - \psi_n^B)}{a}$$

$$\psi_n^B = B \cos(\omega t + \varphi) \cdot \sin(kna)$$

$$\begin{bmatrix} \psi_n^A \\ \psi_n^B \end{bmatrix} = \cos(\omega t + \varphi) \cdot \sin(kna) \begin{bmatrix} A \\ B \end{bmatrix} = e^{i(\omega t + \varphi) - i k n a} \begin{bmatrix} A \\ B \end{bmatrix}$$

$$\psi_{n+1}^A = A e^{i(\omega t + \varphi - kna - ka)} = \psi_n^A / e^{i ka}$$

$$\psi_{n-1}^B = B e^{i(\omega t + \varphi - kna + ka)} = \psi_n^B \cdot e^{i ka}$$

a) Equation of motion

$$\begin{bmatrix} M_A \ddot{\psi}_n^A \\ M_B \ddot{\psi}_n^B \end{bmatrix} = \frac{T_0}{a} \begin{bmatrix} -2 & e^{i ka} + 1 \\ \frac{e^{i ka} + 1}{e^{i ka}} & -2 \end{bmatrix} \begin{bmatrix} \psi_n^A \\ \psi_n^B \end{bmatrix}$$

$$\frac{4M_B}{M_A} \omega^2 \psi_n^A = \frac{T_0}{a} (\psi_n^B (e^{i ka} + 1) - 2 \psi_n^A)$$

$$M_B \omega^2 \psi_n^B = \frac{T_0}{a} (\psi_n^A (\frac{1}{e^{i ka}} + 1) - 2 \psi_n^B)$$

$$\omega^2 \begin{bmatrix} \psi_n^A \\ \psi_n^B \end{bmatrix} = \frac{T_0}{a M_B} \begin{bmatrix} -1/2 & \frac{1+e^{i ka}}{4} \\ 1+e^{-i ka} & -2 \end{bmatrix} \begin{bmatrix} \psi_n^A \\ \psi_n^B \end{bmatrix}$$

$$\omega^2 = \frac{T_0}{a M_B} \lambda^2$$

$$(1 + \cos ka + i \sin ka)(1 + \cos ka - i \sin ka)$$

$$\det(M - \lambda I) = 0 = \begin{vmatrix} -1/2 - \lambda & \frac{1+e^{i ka}}{4} \\ 1+e^{-i ka} & -2 - \lambda \end{vmatrix} = 1 + \frac{5\lambda}{2} + \lambda^2 - \frac{(1+e^{i ka})(1+e^{-i ka})}{4}$$

$$= 1 + \frac{5\lambda}{2} + \lambda^2 - \frac{2 + 2 \overbrace{\cos ka}^{1 - 2 \sin^2 \frac{ka}{2}}}{4}$$

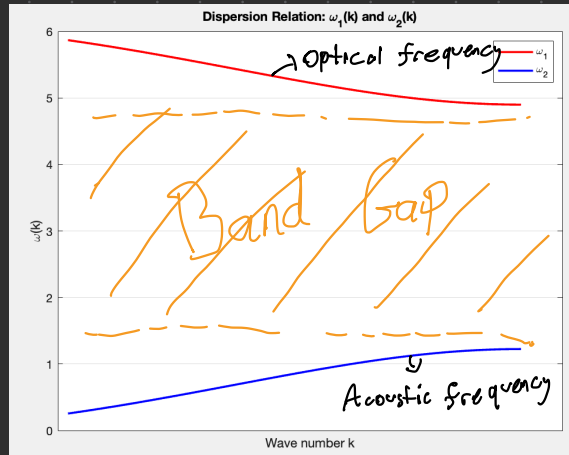
$$= 1 + \frac{5\lambda}{2} + \lambda^2 - \frac{2 + 2 - 4 \sin^2 \frac{ka}{2}}{4} = \sin^2 \frac{ka}{2} + \frac{5\lambda}{2} + \lambda^2$$

$$\lambda = \frac{-\frac{5}{2} \pm \sqrt{\frac{25}{4} - 4 \sin^2 \frac{ka}{2}}}{2}$$

b- Dispersion relation

$$\omega_1 = \sqrt{\frac{T_0}{aM_B}} \cdot \frac{-\frac{\omega}{2} + \sqrt{\frac{25}{4} - 4 \sin^2(\frac{\omega a}{2})}}{2}$$

$$\omega_2 = \sqrt{\frac{T_0}{aM_B}} \cdot \frac{-\frac{\omega}{2} - \sqrt{\frac{25}{4} - 4 \sin^2(\frac{\omega a}{2})}}{2}$$



Even though the eigen values are negative, angular frequencies are positive since $\omega^2 = \frac{T_0}{aM_B} \downarrow$ positive

